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Ex2

write a program to print given number is positive (or) negative

```
n1 = float (input(" enter a number: "))
if (n1 > 0):
    print (n1," +ve number")
else :
    print (n1, "-ve number")
```

Continueous if else (syntax)

```
if (cond1):
    statements of cond1
else :
    if (cond 2) :
        statements of cond2
    else :
        if (cond 3) :
            statements of cond 3
            ....
        else :
            default statements.
```

Syntax if else

```
if (cond1):
    statements of cond1
elif (cond2):
    statements of cond 2
elif( cond 3):
    statements of cond3
.....
else :
    default statements
```

Ex:

```
n1 = float(input("enter a number :"))
if (n1>0):
    print(n1, "+ve number)
elif (n1<0):
    print(n1, "-ve number)
else:
    print("zero")
```

write a program based on percentage display Results

Q:

85 to 100 is distinction
60 to 84 is first class
50 to 59 is second class
35 to 49 is pass
<35 is fail

Execution

```
per = float(input("enter your percentage":))
If (per>=85 and per<=100):
    print("distinction")
elif (per>=60 and per<=84):
    print("first class")
elif (per>=50 and per<=59):
    print("second class")
elif (per>=35 and per<=49):
    print("pass")
elif (per>=0 and <=35):
else:
    print("fail")
print("fail")
else:
    print("Invalid Input")
```

Ex:

check the given inputs that they form which type of triangle

equilateral

Isosceles

Scalene

Execution:

```
a = float( input(" enter a value of 1st side of triangle: "))
b = float (input (" enter a value of 2nd side of triangle: "))
c = float (input (" enter a value of 3rd side of triangle: "))
if (a+b<c (or) b+c<a or c+a<b):
elif (a==b==c): print (" its not a triangle")
    print(" its equilateral triangle")
elif (a==b or b==c or c==a):
    print (" its Isosceles triangle")
else:
    print (" its scalene triangle")
```

Ex:

Electricity Bill calculator

An electricity company charges it's customers as follow:

First 100 units : $0.5 per unit

Next 100 units (101-200): $0.75 per unit

Above 200 units : $1 per unit

Execution:

```
units = int (input (" enter number units used: "))
  if (units>0 and units<=100):
    ta = units * 0.5
  elif (units > 100 and units <=200):
    ta = (100 * 0.5) + ((units - 100) * 0.75)
  elif (units >200):
    ta = (100 * 0.5) + (100 * 0.75) + ((units - 200) * 1)
print (" the total amount is, ta)
```

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Nested conditional statements syntax:

```
If ( condition1 ):
    If ( condition2 ):
        statements of condition2 if
    else :
        statements of condition2 else
else:
    statements of condition1 else
```

BU:

```
# write a program to check positive or negetive
```

Execution

```
num = float (input (" enter a number : "))
If (num != 0):
    if (num > 0):
        print (num, " is +ve number")
    else:
        print (num, " is -ve number")
else:
    print (" zero")
```

AU

```
# ATM withdrawal check, account balance, Amount to withdraw
```

Execution:

```
ba = float( input (" enter balance amount : "))
wa = float(input (" enter withdraw amount : "))
If ( ba > wa):
    If (wa % 100 == 0):
        print (" transaction succeesful")
    else:
        print(" is enter Amount in multiples of 100")
else:
    print(" insufficient balance")
```

Question

Discount calculation, take the total bill amount as input
if bill >= 5000, if bill >= 10000 give 20% discount else give 10% discount
else if bill >= 2000 give 5% discount else no discount

Execution:

```
bill = float(input("enter total bill amount :"))
if bill >= 5000:
    if bill >= 10000:
        print("you got 20% discount")
        da = bill * 20/100
    else:
        print("you got 10% discount")
        da = bill * 10/100
elif bill >= 2000:
    print("you got 5% discount")
    da = bill * 5/100
else:
    print("no diccount")
    da = 0
print(f"the discount amount is : {da}/n the total bill amount is : {bill - da}")
```

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Loops :-

A loop is a programming concept used to execute a block of code repetedly until certain condition is met or the sequence is complete.

Types of loops

* for loop
  1) for loop with sequence
  2) for loop with Range